

NOVA PROGRAM

Nova Charging and Power Conversion Requirements

Domain: Charging

Revision A | Synthetic extraction benchmark

2. Functional requirements

REQ-NVA-CHG-001 | AC input range

The onboard charger shall deliver rated power from 180 V AC through 264 V AC at 45 Hz through 65 Hz.

REQ-NVA-CHG-002 | Conversion efficiency

At 400 V battery voltage and 11 kW output, charger efficiency shall be at least 94 percent.

REQ-NVA-CHG-003 | Power factor

At loads from 50 percent to rated output, input power factor shall be at least 0.99.

3. Verification-oriented requirements

Identifier	Requirement title	Normative requirement
REQ-NVA-CHG-004	Connector temperature	During a 200 A DC charging session, no connector terminal shall exceed 90 deg C, the left-to-right terminal spread shall remain within 8 deg C, and charging shall continue for 20 minutes.
REQ-NVA-CHG-005	Isolation-fault response	When charging isolation falls below 100 kohm, the charger shall stop transferring energy within 100 ms, open the input relays, and report diagnostic CHG-ISO-01.

Applicability notes

Context	Interpretation
Unless a requirement states otherwise	Limits apply to the complete production-intent assembly.
Use of shall	The complete sentence is normative; table columns do not split obligations.

4. Normative installation figure

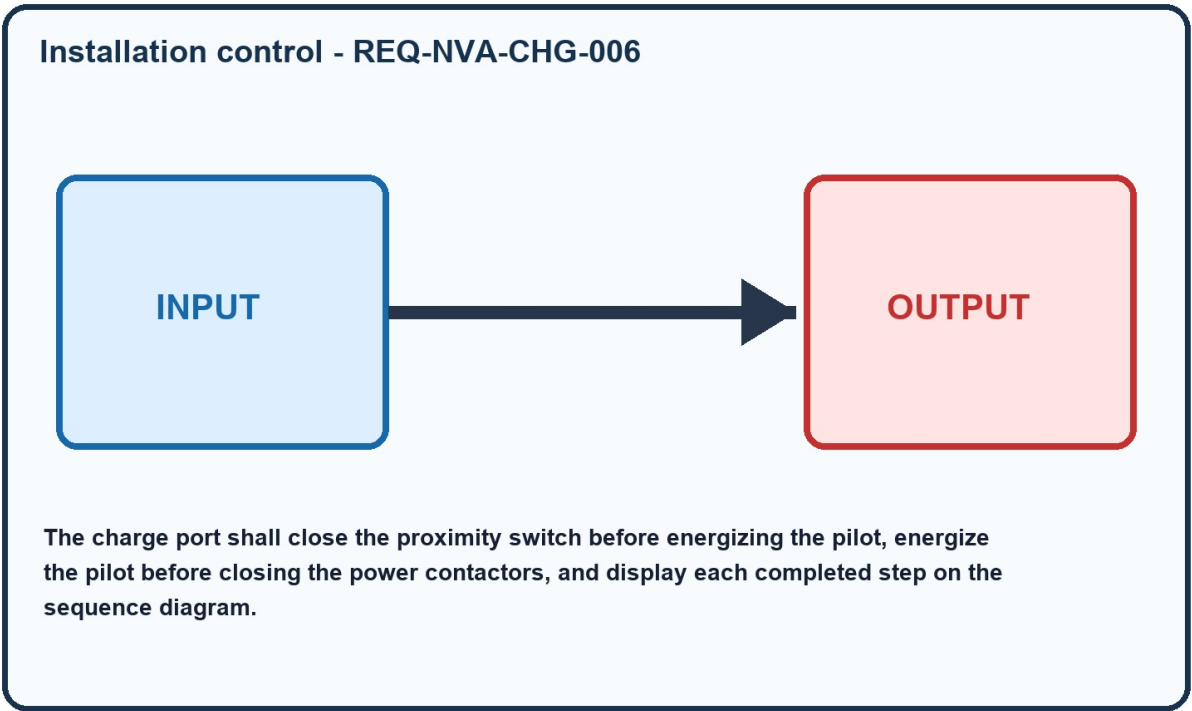


Figure 4-1. Colored geometry and embedded annotation are normative.

Interface context (informative)

